

Global economic connections increase returns to mitigation efforts

Francesco Granella*, Pietro Andreoni†

May 14, 2026

PRELIMINARY WORK IN PROGRESS DO NOT CIRCULATE

Abstract

Economic damages from climate change are commonly estimated by linking local economic outcomes to local weather variation. This approach implicitly treats climate impacts as geographically contained phenomena. Here, we show that climate damages instead emerge from global economic coupling, as temperature shocks propagate across national borders through international trade networks. Exploiting the imperfect correlation of weather shocks across countries and fixed historical trade linkages, we isolate the economic effects of foreign climate exposure independently of local climate conditions. We find that temperature shocks affecting trading partners have larger effects on domestic economic growth than local temperature shocks themselves. Accounting for these cross-border effects in an integrated assessment model substantially increases estimated global economic damages from warming. We show that global economic interdependence supports the case for both unilateral climate action and international cooperation in mitigation efforts.

Keywords: Climate change, climate damages, trade, mitigation.

Climate change is a global physical process expected to cause large welfare losses to most economies in the remaining decades of the century (Dell et al., 2014; Hsiang, 2025). Yet climate change will not affect all regions uniformly. Temperature increases are projected to be larger in polar regions and over land masses, while some regions may experience modest warming or even cooling in specific seasons. Precipitation patterns will shift in complex ways, with some areas becoming wetter while others face increased drought risk. This geographic heterogeneity in climate impacts, combined with variation in local economic structure, natural resource endowments, and adaptive capacity, creates differential climate vulnerabilities across the globe.

Akin to the global climate, the economy is a globally interconnected system, with production, consumption, and investment decisions linked across borders through trade, capital flows, and supply chains (Feng et al., 2024). Like teleconnections in the climate system, these international economic linkages create channels for economic disturbances to propagate globally (Acemoglu et al., 2012). Economic shocks, including those originating from a shifting climate and more frequent or more severe extreme events, are not confined within borders but spill over across networks of trade and international finance (Feng et al., 2024). These long distance linkages raise the question of whether climate damages are best understood as local phenomena or as emergent properties of a globally interconnected economic system.

Recent empirical studies have made significant progress in quantifying the direct effects of local climate on economic activity, documenting substantial nonlinear effects of temperature on economic growth (see Hsiang (2025); Dell et al. (2012); Burke et al. (2015)). This approach largely treats climate change as a local shock, an assumption that is increasingly difficult to reconcile with the structure of the global economy (Astier et al., 2026). On the other hand, model-based estimates rely on descriptions of the world that, although detailed in many respects, are sensitive to economic model structure and parameterization choices (Lemoine et al., 2025). Overlooking the international transmission of climate shocks may lead to systematic misestimation of future climate impacts on the global economy. To address this gap, we quantify the contribution of trade-mediated climate shocks to economic growth and climate damage projections.

*RFF-CMCC European Institute on Economics and the Environment; Euro-Mediterranean Centre on Climate Change.

†Politecnico di Milano; RFF-CMCC European Institute on Economics and the Environment; Euro-Mediterranean Centre on Climate Change.

Exposure to foreign climate shocks

International trade provides a natural conduit for long-distance transmission of shocks. Climate-induced disruptions to production, infrastructure, or labor productivity in one country can affect input availability, prices, and demand conditions in its trading partners (Verschuur et al., 2023). These effects need not be synchronized with local weather and may persist even when domestic climate conditions are benign. Because trade relationships are structured and persistent, they create systematic patterns of exposure to foreign climate shocks that differ across countries. Economies that trade predominantly with warm partners may face higher effective climate exposure than suggested by their own local temperatures, while those integrated with cooler regions may initially experience attenuated impacts.

We therefore construct a measure of weather exposure that combines local weather conditions with those of foreign trading partners. The latter are weighted reflecting historical bilateral trade linkages from BACI (Gaulier and Zignago, 2010) and the openness of the local economy. To limit concerns about endogeneity arising from trade adjustments to contemporaneous climate shocks, we fix weights to their pre-sample values throughout the sample period. This approach captures exogenous variation in foreign climate exposure driven by weather fluctuations in partner countries, while isolating the influence of trade-mediated transmission from changes in trade patterns induced by climate change itself. Weights do not correlate with pairwise country temperature correlations, and do not reproduce teleconnections in the climate system (Figure 2). Combined with climate data from ERA5 Land, we build a global panel of countries over the period 1996-2024.

Main results

Using an established panel fixed effects regression (Burke et al., 2015, 2026), we find that trade-weighted temperatures exert an economically meaningful and nonlinear effects on domestic economic growth (Table 1, Column 1). The effect is substantially larger than those of contemporaneous local temperature shocks alone (Column 2), meaning that a substantial share of climate damages experienced by national economies originates abroad.

A uniform $+1^{\circ}\text{C}$ increase in global temperatures at recent income and climate levels reduces yearly global GDP growth by 0.08 percentage points per year. When foreign exposure is ignored, the same warming yields a reduction of 0.04 percentage points. Because trade-mediated exposure amplifies the response of economic growth to warming, damages compound across countries rather than remaining confined within national borders. As a result, aggregate losses rise disproportionately relative to projections based solely on local climate impacts.

Model fit statistics further support the empirical relevance of foreign climate exposure. Relative to the local-only specification (Column 2), which restricts identification to within-country weather variation, the baseline model including trade-weighted temperature (Column 1) has a higher within adjusted R^2 (0.0016 against 0.0001) and a lower BIC ($-17, 207.90$ against $-17, 201.65$). Because all specifications share the same country fixed effects, year fixed effects, and country-specific linear trends, these gains reflect explanatory power attributable specifically to the trade-mediated component of climate exposure, net of any absorbed by the fixed-effect structure. The data therefore favor a specification in which domestic economic performance responds to both local and trade-weighted weather, consistent with the theoretical premise that climate shocks propagate through international trade linkages rather than remaining confined within national borders.

Global temperature fluctuations have been shown to predict local economic growth more strongly than local weather alone, raising questions about the channels through which climate affects economic outcomes (Bilal and Känzig, 2024). One interpretation is that extreme atmospheric events correlated at the global scale drive these effects. An alternative explanation, which we investigate here, is that climate damages propagate through economic networks, so that foreign climate shocks directly shape domestic economic performance. Under this view, global temperature measures proxy for the cumulative effect of geographically dispersed but economically connected climate shocks.

To corroborate our hypothesis that damages are transmitted through trade linkages, we assign equal weights to trade partners in 1995 and re-construct trade weighted temperatures. If the results are driven by global climate shocks, then the geography of trade networks would not matter for trade-mediated exposure to foreign climate shocks and trade-weighted temperatures in our placebo regression should have a significant effect on GDP growth. Column 3 of Table 1 shows that uniformly weighted foreign temperatures have no statistically significant effect on economic growth, suggesting that the main results are not driven by spurious correlations, general climatic patterns, or large extreme weather events. This further confirms that the approach of Bilal and Känzig (2024) implicitly capture the effect of weather shocks being transmitted through trade networks.

We next assess whether the main result is sensitive to the sectoral composition of the trade weights and to the underlying bilateral trade dataset. Columns 4 and 5 of Table 1 reconstruct foreign exposure using only agricultural trade flows and only industrial trade flows, respectively. The estimated coefficients on $\tilde{T}$ and $\tilde{T}^2$ are essentially indistinguishable from the baseline in Column 1 (0.0126 and 0.0129 on the linear term, against 0.0126 in the baseline), indicating that transmission is not confined to a single sector but is a general feature of goods trade. Columns 6 and 7 replace the harmonized BACI weights with those derived from raw COMTRADE and from the longer-historical TRADHIST database. Despite substantially smaller samples (2,611 and 2,830 observations, against 4,144 in the baseline) reflecting the narrower country–year coverage of the two alternatives, the nonlinear trade-weighted temperature response remains concave, statistically significant, and of the same order of magnitude as in Column 1, with point estimates that are if anything somewhat larger. Taken together, these four specifications indicate that the transmission of foreign climate shocks documented in Column 1 is robust to the sector used to measure trade linkages and to the specific data source chosen to construct the weights.

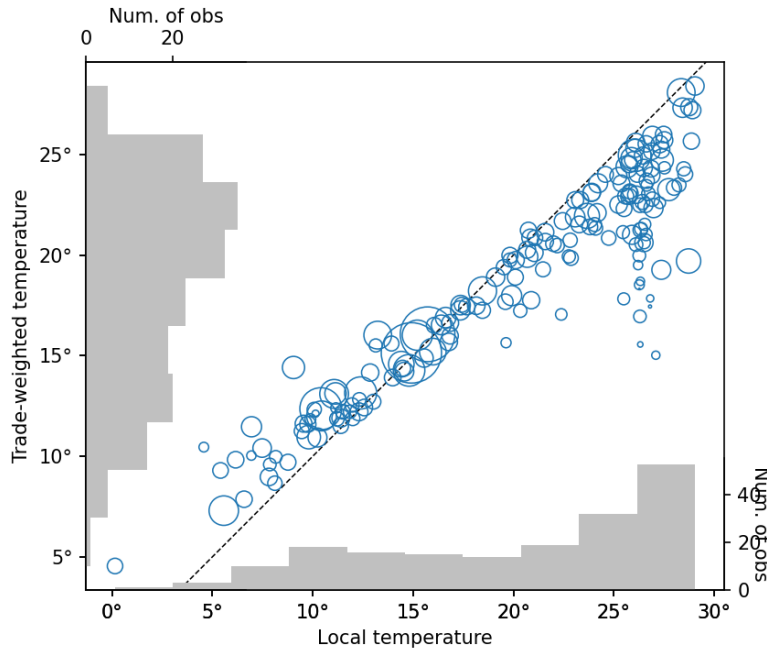


Figure 1: **Trade-weighted climate exposure differs systematically from local climate, reflecting global economic coupling.** Local temperature against trade-weighted temperature. The size of the circles is proportional to the average trade weights of each country in 2015-2019. The histogram represents the unweighted distribution of local temperatures (right axis). The dashed line traces the equality between the two measures.

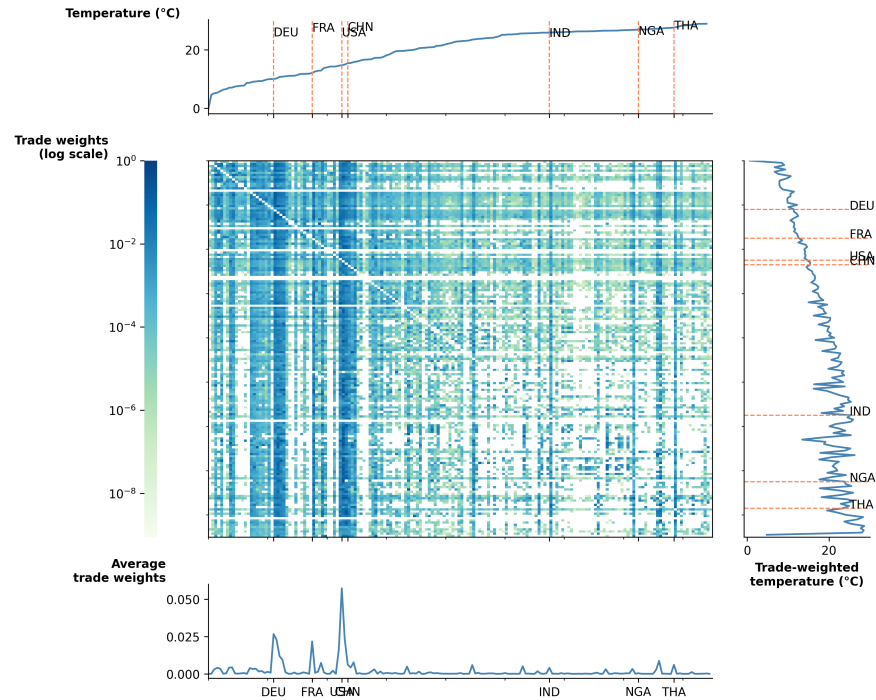


Figure 2: **Trade weights, local temperatures, and trade-weighted temperatures.** The central panel displays the bilateral trade weights averaged over 2010 to 2019, with countries on both axes sorted by annual mean temperature, shown in the upper panel. The right panel displays the trade-weighted temperature, inclusive of self-trade. The bottom panel displays the average trade weight that each country has in the data.

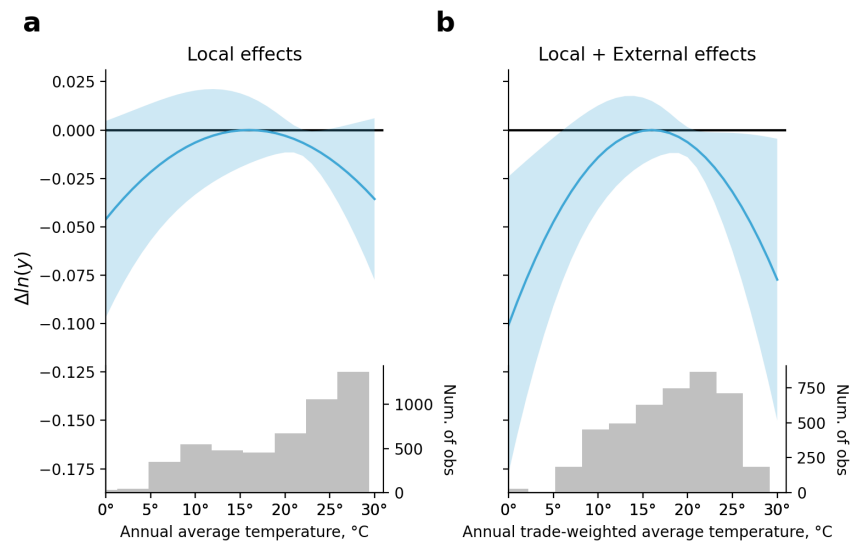


Figure 3: Effects on GDP per capita growth of local and external weather shocks.

Figure 4: **Notes here.** More notes here.

Table 1: The effects of weather shocks on local economic performance.

	Trade	Local only	Uniform weights	Agriculture	Industry	COMTRADE	TRADHIST
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	$\Delta \ln(y)$	$\Delta \ln(y)$	$\Delta \ln(y)$	$\Delta \ln(y)$	$\Delta \ln(y)$	$\Delta \ln(y)$	$\Delta \ln(y)$
$\tilde{T}$	0.0126** (0.005)	0.0058** (0.003)	-0.0001 (0.000)	0.0126** (0.005)	0.0129** (0.005)	0.0183*** (0.005)	0.0151*** (0.005)
$\tilde{T}^2$	-0.0004** (0.000)	-0.0002* (0.000)	0.0000 (0.000)	-0.0004** (0.000)	-0.0004** (0.000)	-0.0006*** (0.000)	-0.0005*** (0.000)
$\tilde{P}$	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Country FE	Yes	Yes	No	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes	Yes
Country-Year trends	Yes	Yes	No	Yes	Yes	Yes	Yes
N	4144	4144	4144	4144	4144	2611	2830
Adj-R ₂	0.3606	0.3596	0.3591	0.3605	0.3606	0.3623	0.4111
Within adj-R ₂	0.0016	0.0001	-0.0008	0.0015	0.0017	0.0035	0.0036
BIC	-17207.8964	-17201.6462	-17198.0082	-17207.1687	-17208.2194	-11413.4335	-11676.8078

Notes: $\tilde{T}$ is trade-weighted using 1995 volumes from BACI. In all regressions the dependent variable is the growth rate of GDP per capita, and use post-1995 data. Standard errors clustered by country in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Global economic interconnectedness raises the case for both unilateral action and international cooperation

An upward revision of projected climate damages supports the case for unilateral climate action even when evaluated from a domestic perspective (Bilal and Känzig, 2024). We augment the RICE50x integrated assessment model (Gazzotti, 2022) with a gravity equation to project future trade networks and with a damage function that incorporates foreign climate shocks. We project local and trade-related shocks separately, and derive the implications for optimal unilateral abatement. We find that including internationally transmitted climate shocks increases the net present value of avoided damages from domestic mitigation efforts, reflecting the additional domestic benefits arising from reduced exposure to foreign climate shocks when global temperatures are lowered through local abatement (Figure ??).

Optimal abatement rates are in fact higher once the international transmission of climate shocks through trade networks is taken into account (Figure ??). In addition, the international transmission of climate shocks through trade networks creates structural interdependencies that enhance the benefits of cooperative climate policies, supporting the case for global cooperation in mitigation and adaptation efforts.

Discussion

This study shows that climate damages are not confined to the locations where climate shocks occur, but instead propagate through global economic linkages. By quantifying the transmission of temperature shocks through international trade networks, we demonstrate that a substantial share of the economic impacts of climate change experienced by national economies originates abroad rather than domestically. As a result, estimates based solely on local climate variation systematically understate both the magnitude and the geographic concentration of climate damages.

These results have implications for how climate damages are measured and projected. Empirical approaches that rely exclusively on within-country weather variation implicitly treat climate shocks as independent across space and therefore miss an important component of economic exposure. Incorporating cross-border transmission alters both short-run estimates and long-run projections, leading to higher global damages and greater inequality than previously recognized. Integrated assessment models that abstract from international transmission mechanisms are therefore likely to underestimate the economic risks associated with climate change, particularly for highly interconnected economies.

The analysis also reframes the role of international economic integration in shaping climate risk. While trade can facilitate adaptation by reallocating resources in response to local shocks (Costinot et al., 2016; Janssens et al., 2020), it simultaneously

acts as a conduit for the propagation of climate damages. Greater economic integration increases exposure to foreign climate conditions, linking domestic economic outcomes to climate shocks occurring elsewhere. As global supply chains deepen, this form of exposure becomes increasingly important for understanding macroeconomic vulnerability to climate change.

It should be noted that it is an established result of economics theory and empirics growing autarchy, while reducing exposure to foreign climate shocks, leads to allocative inefficiencies and forfeits the gains of trade (Arkolakis et al., 2012).

Finally, our findings highlight a structural basis for international climate interdependence. Because economic exposure to climate shocks extends across borders, actions that reduce climate vulnerability in one country can affect economic outcomes in its trading partners. This interdependence implies that the economic benefits of climate mitigation and adaptation are not confined within national boundaries, even when evaluated from a narrow domestic perspective. Accounting for trade-mediated climate transmission therefore changes how the costs and benefits of climate action are distributed across countries.

More broadly, the results suggest that climate change should be understood not only as a local environmental stressor, but as a systemic risk shaped by the architecture of the global economy. Accurately assessing its economic consequences requires moving beyond locally grounded damage functions toward frameworks that explicitly account for international economic coupling. As climate change intensifies and global integration continues, such system-level approaches will become increasingly important for understanding and managing the full economic impacts of a warming world.

Methods

Empirical strategy

Measuring foreign climate exposure

For each country and year, we construct a trade-weighted weather index that aggregates temperature and precipitation across all trading partners and the local economy. Local weather has a weight proportional to the domestic (trade-independent) economy, measured using the Penn World Table (PWT) 11.0 expenditure shares. Let ϕ_c , ϕ_i , and ϕ_g denote the current-PPP shares of private consumption, investment, and government consumption, respectively, and ϕ_x , ϕ_m , and ϕ_r denote the shares of exports, imports, and the residual trade and statistical discrepancy. We define the trade-independent share as domestic absorption minus imports, $s_i = (\phi_c + \phi_i + \phi_g - \phi_m)_i$. This measure removes the direct contribution of imports from expenditure. The corresponding trade-dependent share is defined as $1 - s_i$, capturing the portion of output associated with traded volumes and the residual balancing term.

Bilateral trade volumes from 1995 determine foreign weights, reflecting the intensity of economic linkages through which climate shocks may propagate. Fixing weights prior to the estimation period limits concerns about endogeneity from contemporaneous trade adjustments. Formally, we build the weather indices as:

$$\tilde{T}_{it} = s_i \cdot T_{it} + (1 - s_i) \sum_{j \neq i} \frac{F_{ij,1995}}{\sum_{j \neq i} F_{ij,1995}} T_{jt}; \quad \tilde{P}_{it} = s_i \cdot P_{it} + (1 - s_i) \sum_{j \neq i} \frac{F_{ij,1995}}{\sum_{j \neq i} F_{ij,1995}} P_{jt}$$

$F_{ij,1995}$ is total bilateral trade (imports plus exports) between i and j in 1995. Data on trade flows comes from BACI (reference).

Climate and economic data

Trade data

Bilateral trade data is sourced from the BACI (Base pour l'Analyse du Commerce International) dataset, constructed by the Centre d'Études Prospectives et d'Informations Internationales (CEPII) (Gaulier and Zignago, 2010). It provides a harmonized version of UN COMTRADE data that addresses several methodological limitations of the original source. BACI employs a reconciliation procedure that exploits the dual reporting of trade flows to produce more accurate bilateral trade estimates by systematically comparing mirror flows and selecting the most reliable values based on data quality indicators. The dataset extends from 1995 to 2024, covering approximately 200 countries with trade flows disaggregated at the 6-digit level of the Harmonized System classification, and includes both nominal values and quantities where available. This reconciliation methodology reduces the measurement error inherent in raw COMTRADE data. We exclude data on 2020 and 2021 due to COVID-19-related

disruptions in the global trade schemes. We aggregate disaggregated flows from BACI, which provides a harmonized and reconciled version of COMTRADE data, to total flows and by macro-sectoral level, agriculture and industry.¹ The sample is an unbalanced panel of 163 countries for 27 years.

Climate data processing

Temperature and precipitation data is obtained from the ERA5-Land hourly dataset, which offers high-resolution climate data (Mu   oz-Sabater et al., 2021). ERA5-Land is a reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF) and provides comprehensive information on climate variables, including temperature and precipitation, at a global scale. The dataset is available at a spatial resolution of 0.25 degrees. We aggregate hourly data to annual means (temperature) and sums (precipitation) at the country-year level using gridded population of 2000 as weights. The outcome of interest is the rate of growth of the GDP per capita in constant 2015 USD, from the World Bank’s World Development Indicators.

Empirical strategy

We estimate the effect of foreign climate exposure on domestic economic growth using a panel regression that includes country fixed effects, year fixed effects, and country-specific linear time trends. This specification absorbs time-invariant country characteristics, common global shocks, and smooth long-run growth trajectories, ensuring that identification comes from local and foreign climate shocks.

Climate effects are modeled using a quadratic temperature response function, allowing for nonlinear impacts at different temperature levels. Call y_{it} the GDP per capita of country i in year t . The estimating equation is:

$$\Delta \ln(y_{it}) = \beta_1 \tilde{T}_{it} + \beta_2 \tilde{T}_{it}^2 + \beta_3 \tilde{P}_{it} + \beta_4 \tilde{P}_{it}^2 + \mu_i + \lambda_t + \delta_i \cdot t + \varepsilon_{it} \quad (1)$$

Standard errors are clustered at the country level to account for serial correlation. Robustness analyses explore alternative trade datasets, weighting schemes, lag structures, and placebo tests, as described in the Supplementary Methods.

Climate damage projections

To assess the implications of trade-mediated climate transmission for future damages, we incorporate the estimated foreign climate effects into the RICE50+ integrated assessment model (Gazzotti, 2022) with global economic and climate dynamics. Trade linkages are reconstructed using a gravity-based framework, allowing foreign climate exposure to evolve endogenously with projected economic development and climate change. We compare projections that include trade-mediated transmission with counterfactual scenarios that account only for local climate impacts.

¹Table ?? describes the mapping from Harmonized System 6-digit classification to macro-sectors. Note that data on cross-country flows of services are not available.

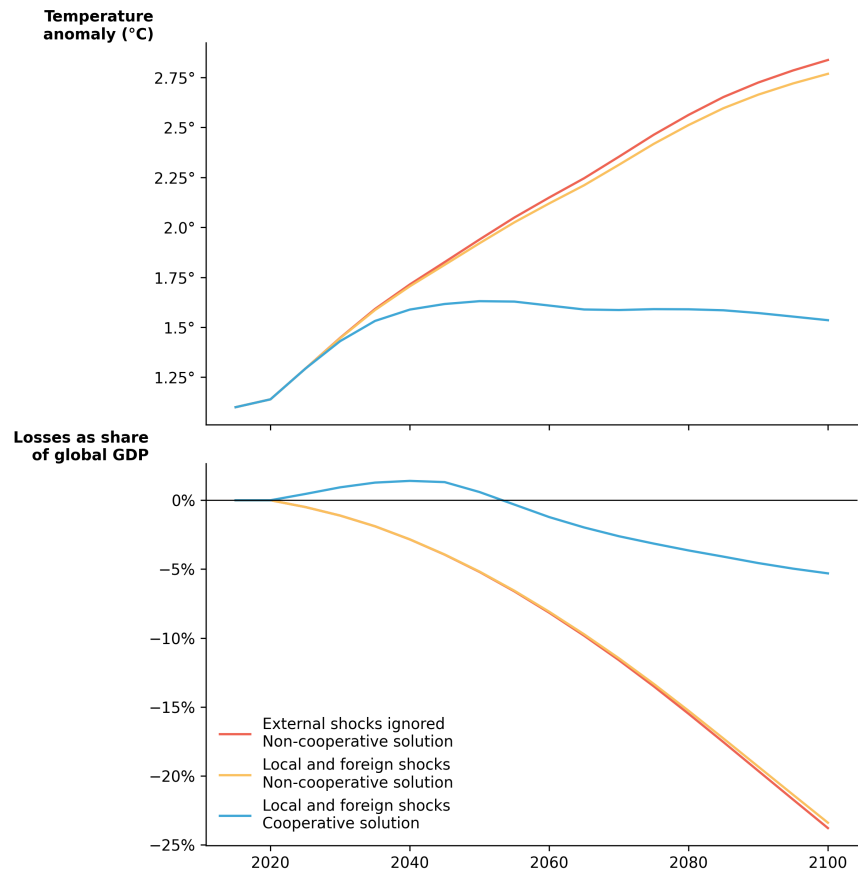


Figure 5: **Notes here.** More notes.

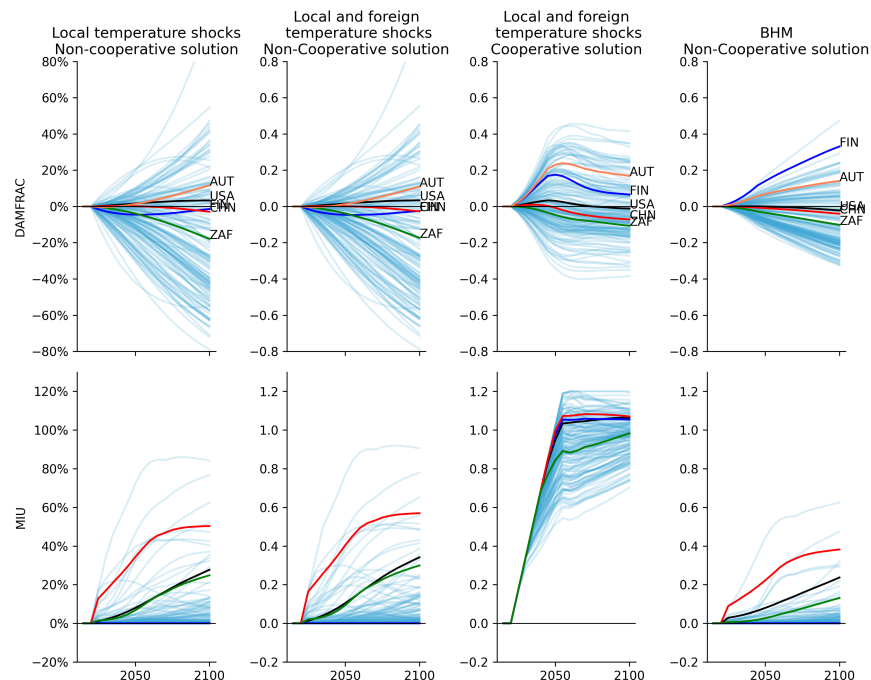


Figure 6: **Notes here.** More notes.

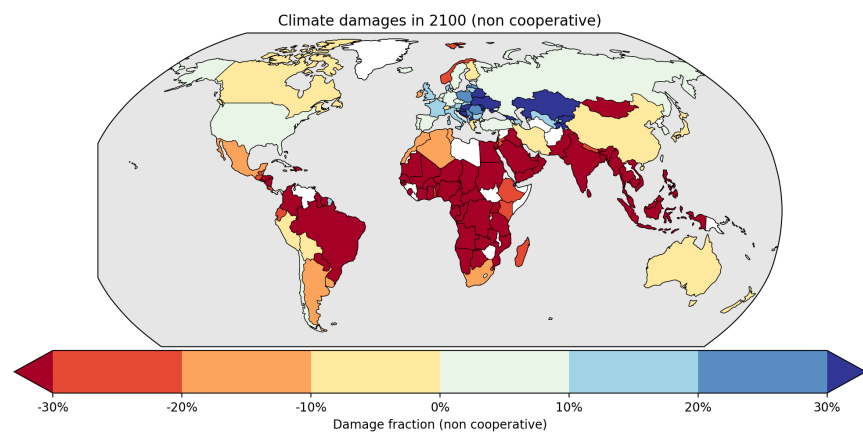


Figure 7: **Notes here.** More notes.

References

- Acemoglu, D., Carvalho, V. M., Ozdaglar, A., and Tahbaz-Salehi, A. (2012). The network origins of aggregate fluctuations. *Econometrica*, 80(5):1977–2016.
- Arkolakis, C., Costinot, A., and Rodr  guez-Clare, A. (2012). New trade models, same old gains? *American Economic Review*, 102(1):94–130.
- Astier, J., Barrows, G., Calel, R., and Ollivier, H. (2026). Quantifying climate damages when regions trade: A structural gravity approach. Technical report.

- Bilal, A. and Känzig, D. R. (2024). The Macroeconomic Impact of Climate Change: Global vs. Local Temperature. Working Paper 32450.
- Burke, M., Hsiang, S. M., and Miguel, E. (2015). Global non-linear effect of temperature on economic production. *Nature*, 527(7577):235–239.
- Burke, M., Zahid, M., Diffenbaugh, N. S., and Hsiang, S. (2026). Quantifying climate loss and damage consistent with a social cost of carbon. *Nature*, 651(8107):959–966.
- Costinot, A., Donaldson, D., and Smith, C. (2016). Evolving comparative advantage and the impact of climate change in agricultural markets: Evidence from 1.7 million fields around the world. *Journal of Political Economy*, 124(1):205–248.
- Dell, M., Jones, B. F., and Olken, B. A. (2012). Temperature Shocks and Economic Growth: Evidence from the Last Half Century. *American Economic Journal: Macroeconomics*, 4(3):66–95.
- Dell, M., Jones, B. F., and Olken, B. A. (2014). What do we learn from the weather? the new climate-economy literature. *Journal of Economic Literature*, 52(3):740–98.
- Feng, A., Li, H., and Wang, Y. (2024). We are all in the same boat: Cross-border spillovers of climate shocks through international trade and supply chains. *The Economic Journal*, 135(669):1433–1466.
- Fouquin, M. and Hugot, J. (2016). Two centuries of bilateral trade and gravity data: 1827-2014. Working Paper 14, Centre d’Études Prospectives et d’Informations Internationales.
- Gaulier, G. and Zignago, S. (2010). Baci: International trade database at the product-level. the 1994-2007 version. Working Papers 2010-23, CEPII.
- Gazzotti, P. (2022). RICE50+: DICE model at country and regional level. *Socio-Environmental Systems Modelling*, 4:18038–18038.
- Hsiang, S. (2025). *The Global Economic Impact of Climate Change: An Empirical Perspective*.
- Janssens, C., Havlíř, P., Krisztin, T., Baker, J., Frank, S., Hasegawa, T., Leclercq, D., Ohrel, S., Ragnauth, S., Schmid, E., Valin, H., Van Lipzig, N., and Maertens, M. (2020). Global hunger and climate change adaptation through international trade. *Nature Climate Change*, 10(9):829–835.
- Lemoine, D., Hausman, C., and Shrader, J. (2025). *A Guide to Climate Damages*.
- Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., Boussetta, S., Choulga, M., Harrigan, S., Hersbach, H., Martens, B., Miralles, D. G., Piles, M., Rodríguez-Fernández, N. J., Zsoter, E., Buontempo, C., and Thépaut, J.-N. (2021). ERA5-land: a state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data*, 13(9):4349–4383.
- Verschuur, J., Koks, E. E., and Hall, J. W. (2023). Systemic risks from climate-related disruptions at ports. *Nature Climate Change*, 13(8):804–806.

A Supplementary Methods

A.1 Supplementary figures

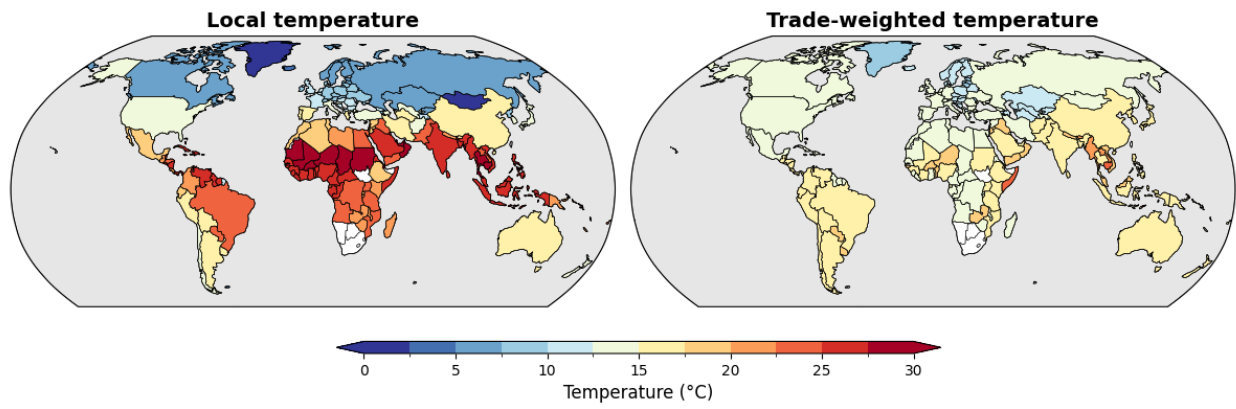


Figure 8: Trade-weighted climate exposure differs systematically from local climate, reflecting global economic coupling.

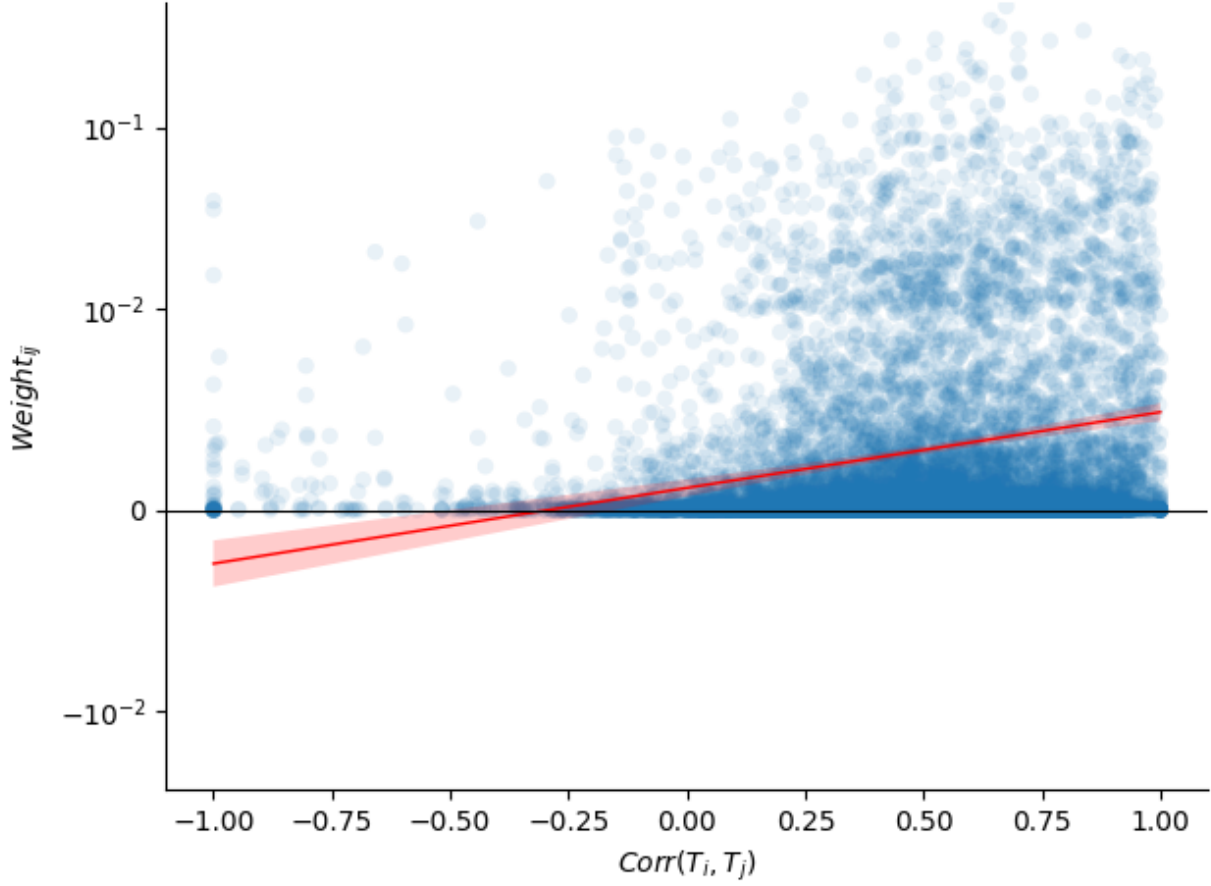


Figure 9: Pairwise correlations of local temperatures between country pairs (1995-2024) do not meaningfully predict pre-sample trade weights.

A.2 Summary statistics

Table 2: Summary statistics.

	Mean	S.D.	Min.	Max.
$\Delta \ln(y)$	0.02	0.04	-0.20	0.19
T	19.54	7.23	-1.80	29.37
P	1139.35	730.25	7.56	5159.52
$\tilde{T}$	18.03	5.70	-0.75	29.18
$\tilde{P}$	1140.43	579.49	112.68	4398.00
$\frac{Trade}{GDP}$	0.53	0.30	0.08	2.28
Observations	4144			

Notes: Summary statistics for the main sample used in the regressions. Data span the period 1996-2019. $\Delta \ln(y)$ is the annual GDP per capita growth rate. T population-weighted country-level annual average temperature expressed in degree Celsius. P is the population-weighted country-level annual total precipitation, in millimeters. Trade flows-weighted temperature $\tilde{T}_{(\cdot)}$ and precipitation $\tilde{P}_{(\cdot)}$ are computed using BACI trade data for the year 1995.

A.3 Robustness

A.3.1 Alternative trade datasets

To validate our findings, I conduct several robustness checks. I first test whether the results are dependent on the choice of the dataset on historical trade volumes. Depending on the data source, coverage across countries and over time may vary. I estimate Equation 1 for two alternative datasets, TRADHIST and COMTRADE. To avoid endogeneity issues arising from contemporaneous trade flows, I exclude the first year of the time series from the estimation of Equation 1 and use its trade volumes as fixed weights for calculating foreign temperature shocks.

The TRADHIST database (Fouquin and Hugot, 2016) which provides 1.9 million bilateral trade observations for the 1827-2014 period. The dataset exclusively uses nominal series to maintain consistency with structural gravity theory requirements. The dataset was specifically designed to analyze the two major waves of globalization, the First Globalization of the 19th century and the post-World War II Second Globalization, the second one being the focus of this paper. I focus on post-1995 years for consistency with BACI, and to better reflect recent and near-future trade patterns.

I also test the United Nations Commodity Trade Statistics Database (COMTRADE) which BACI builds upon. COMTRADE contains bilateral trade flows reported by national statistical offices to the UN Statistics Division. The database covers traded goods between over 200 countries and territories, classified according to standardized commodity nomenclatures including the Harmonized System (HS) and Standard International Trade Classification (SITC), with annual data supplemented by monthly statistics for major trading nations. While COMTRADE's extensive coverage and temporal depth make it indispensable for empirical trade research, users must account for well-documented limitations including reporting lags, inconsistencies between mirror flows, and varying data quality across countries and time periods. I use trade flows as reported by the importing countries, which are generally considered more reliable than those reported by exporters due to incentives for accurate reporting of goods entering the economy. I focus on post-1995 years for comparability with the main results.

Results using both TRADHIST (Column 1) and COMTRADE (Column 2) are qualitatively consistent with those obtained using the harmonized BACI (Table 3). However, local weather shocks have no significant effect, whereas the effects of foreign weather shocks remain large, concave, and statistically significant.

A.3.2 Ruling out the effects of global climate shocks

Table 3: Regression results using alterative weighting schemes.

	(1)	(2)	(3)
	TRADHIST	COMTRADE	Placebo
T	0.0002 (0.003)	-0.0025 (0.004)	0.0056** (0.002)
T^2	-0.0002* (0.000)	-0.0001 (0.000)	-0.0002*** (0.000)
$\tilde{T}_T$	0.0497*** (0.012)	0.0599*** (0.015)	-0.0007 (0.002)
$\tilde{T}_T^2$	-0.0017*** (0.000)	-0.0021*** (0.001)	0.0000 (0.000)
$P, \tilde{P}$	Yes	Yes	Yes
Country FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Country-Year trends	Yes	Yes	Yes
N	3310	2683	4643
Adj-R ²	0.4008	0.3651	0.3481
Within adj-R ²	0.0077	0.0129	-0.0001

Notes: In all regressions the dependent variable is the growth rate of GDP per capita. Column 1 and 2 use TRADHIST and COMTRADE datasets for trade weights, where weights are total trade (the sum of imports and exports) volumes in 1995. All regressions include precipitation, squared precipitation, and their trade-weighted equivalents, and use only post-1995 data. Standard errors clustered by country in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. All regressions include precipitation, squared precipitation, and their trade-weighted equivalents. Standard errors clustered by country in parenthesis. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.